1) If function f and g are such that $f(x) = g(x) + k$, Where k is a constant

a. $f'(x) = g'(x) + k$

b. $f'(x) = g'(x)$

c. $f''(x) = g''(x)$

d. both b and c

2) If function $y(x) = 5x^4 + 3x^2 + 2x + 6$, then $y''(x) =$

a. $20x^3 + 6x + 2$

b. $80x^3 + 12$

c. $60x^2 + 6$

d. $120x$

3) Find the integral

$$\int_2^3 (8x^3 + 3x^2 + 6)\, dx$$

a. 88.

b. 155

c. $8x^4$

d. $6x^4$

4) Solve $8x + 7y = 38$ and $3x - 5y = -1$

a. No solution.

b. Infinite solution.

c. Unique solution

d. None of the above

5) Find the area of a triangle having base of 8 cm and height of 4 cm.

a. 16 square cm
b. 32 square cm
c. 8 square cm
d. 64 square cm

6) Economics deals primarily with the concept of

a. scarcity
b. money
c. poverty
d. banking

7) If the cost of printer increases, what will be the effect on demand of cartridge?

a. Increase
b. Decrease
c. No effect
d. None of the above

8) The science of economics is related to

a. Human behaviour.

b. Political rights.

c. Welfare.

d. Nation's history.

9) If the cost of chicken increases, what will be the effect on demand of fish?

a. Increase
b. Decrease
c. No effect
d. None of the above

10) A large increase in the number of fast-food restaurants in a community is most likely to result in

a. Lower prices and higher quality.
b. Lower prices and lower quality.
c. Higher prices and higher quality.
d. None of the above.

9) If